SHEET2

SIC Machine

1. Translate (by hand) the following assembly program to SIC object code. Which contains H record, T record, and E record.

Example (1):

```
Prog1     Start     0000
          LDA       ZERO
          STA       INDEX
LOOP      LDX       INDEX
          LDA       ZERO
          STA       ALPHA, X
          LDA       INDEX
          ADD       THREE
          STA       INDEX
          COMP      K300
          TIX       TWENTY
          JLT       LOOP
INDEX     RESW      1
ALPHA     RESW      100
ZERO      WORD      0
K300      WORD      100
THREE     WORD      3
TWENTY    WORD      20
End       0000
```

Example (2):

```
HW1A      START     1010
LDA       ALPHA
STA       GAMMA
LDA       BETA
STA       ALPHA
LDA       GAMMA
STA       BETA
ALPHA     WORD      100
BETA      WORD      333
GAMMA     RESW      1
End       1010
```

Example (3):

```
HW1B     START    0030
WRREC    LDX      ZERO
WLOOP    TD       OUTPUT
         JEQ      WLOOP
         LDCH     RECORD, X
         WD       OUTPUT
         TIX      LENGTH
         JLT      WLOOP
ZERO     WORD     0
LENGTH   WORD     1
OUTPUT   BYTE     X '05'
RECORD   RESB     100
End      0030
```

Example (4):

```
COPY      Start    1000
FIRST     STL      RETADR
CLOOP     JSUB     RDREC
          LDA      LENGTH
          COMP     ZERO
          JEQ      ENDFIL
          JSUB     WRREC
          J        CLOOP
          LDA      EOF
          STA      BUFFER
ENDFIL    LDA      THREE
          STA      LENGTH
          JSUB     WRREC
          LDL      RETADR
          RSUB
EOF       BYTE     C'EOF'
THREE     WORD     3
ZERO      WORD     0
RETADR    RESW     1
LENGTH    RESW     1
BUFFER    RESB     4096
RDREC     LDX      ZERO
          LDA      ZERO
RLOOP     TD       INPUT
          JEQ      RLOOP
          RD       INPUT
          COMP     ZERO
```

```
        JEQ     EXIT
        STCH    BUFFER
        TIX     MAXLEN
        JLT     RLOOP
EXIT    STX     LENGTH
        RSUB
INPUT   BYTE    X'F1'
MAXLEN  WORD    4096
WRREC   LDX     ZERO
WLOOP   TD      OUTPUT
        JEQ     WLOOP
        LDCH    BUFFER,X
        WD      OUTPU

        TIX     LENGTH
        JLT     WLOOP
OUTPUT  RSUB
        BYTE    X'05'
        END
```